

APB4 INTERFACE SIGNALS & FSM NOTES

Based on ARM IHI 0024E - AMBA APB Protocol Specification

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Personal study note for APB4 / RTL / Design Verification

Note: I rewrote these notes in my own words so I can understand APB4 more easily. The writing is intentionally simple, like a Vietnamese engineering student preparing for an internship interview. The technical information is taken from the APB specification, especially Appendix A, Appendix B, Chapter 2, and Chapter 4.

1. APB4 Signal List

First, I separate the signals into three groups: Requester-driven (Master -> Slave), Completer-driven (Slave -> Master), and common clock/reset signals. In the newer ARM specification, the words Requester and Completer are used. For me, I still connect them with Master and Slave because this is easier to remember from older APB diagrams.

Table B-1 code meaning: Y = mandatory, O = optional, and OO = optional for an output port but mandatory for an input port.

1.1 Requester-driven signals (Master -> Slave)

Signal	APB4	Width	Meaning	Validity condition
PADDR	Y	ADDR_WIDTH (up to 32 bits)	Address of the transfer	Valid when PSELx = 1
PPROT	O	3 bits	Protection attributes	Valid when PSELx = 1, if implemented
PSELx	Y	1 bit per Completer	Selects one peripheral	Always valid
PENABLE	Y	1 bit	Marks ACCESS phase	Valid when PSELx = 1
PWRITE	Y	1 bit	1 = write, 0 = read	Valid when PSELx = 1
PWDATA	Y	DATA_WIDTH (8/16/32 bits)	Write data	Active write lanes valid when PSELx = 1
PSTRB	O	DATA_WIDTH/8	Byte write strobe	Valid when PSELx = 1, if implemented

Small note about PSTRB: there is one strobe bit for each byte of PWDATA. During a read transfer, PSTRB should be all LOW. I think this is an easy detail to forget in verification.

1.2 Completer-driven signals (Slave -> Master)

Signal	APB4	Width	Meaning	Validity condition
PREADY	OO	1 bit	Completer says the transfer can finish / can insert wait states	Valid when PSELx = 1 and PENABLE = 1
PRDATA	Y	DATA_WIDTH (8/16/32 bits)	Read data	Valid at completion of a read: PSELx=1, PENABLE=1, PREADY=1
PSLVERR	OO	1 bit	Transfer error response	Valid only at completion: PSELx=1, PENABLE=1, PREADY=1

Why OO is important: OO is not the same as O. For example, a slave side can treat PREADY as a simple constant HIGH when it never needs wait states, while an input that expects PREADY still needs a defined value. Table B-1 gives the default PREADY = 1 and PSLVERR = 0.

1.3 Clock and Reset

Signal	APB4	Width	Meaning	Validity condition
PCLK	Y	1 bit	Clock	All APB signals are timed with the rising edge of PCLK
PRESETn	Y	1 bit	Active-low reset	Normally connected to the system bus reset

Important: PCLK and PRESETn are mandatory APB4 signals, but I do not classify them as Requester-driven or Completer-driven because they come from the clock/reset system.

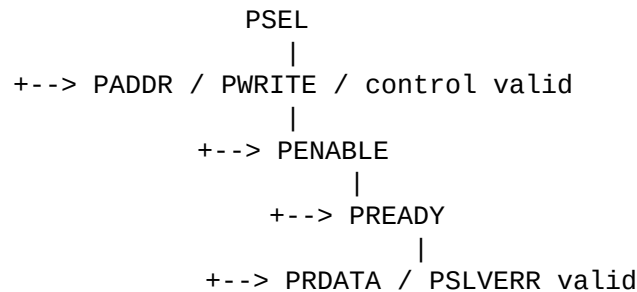
2. Appendix A - Signal Validity Rules

For APB4, I can remember signal validity in levels. Each next level means the transfer is closer to completion.

Condition	Signals that must be valid
Always	PSELx
PSELx = 1	PADDR, PPROT* , PENABLE, PWRITE, PSTRB* , PWDATA active write lanes
PSELx = 1 and PENABLE = 1	PREADY
PSELx = 1 and PENABLE = 1 and PREADY = 1	PRDATA for read, PSLVERR

* *Optional APB4 signal.* The rule applies if that signal exists on the interface.

My easiest memory chain is: PSEL -> request information valid -> PENABLE -> PREADY -> response valid.



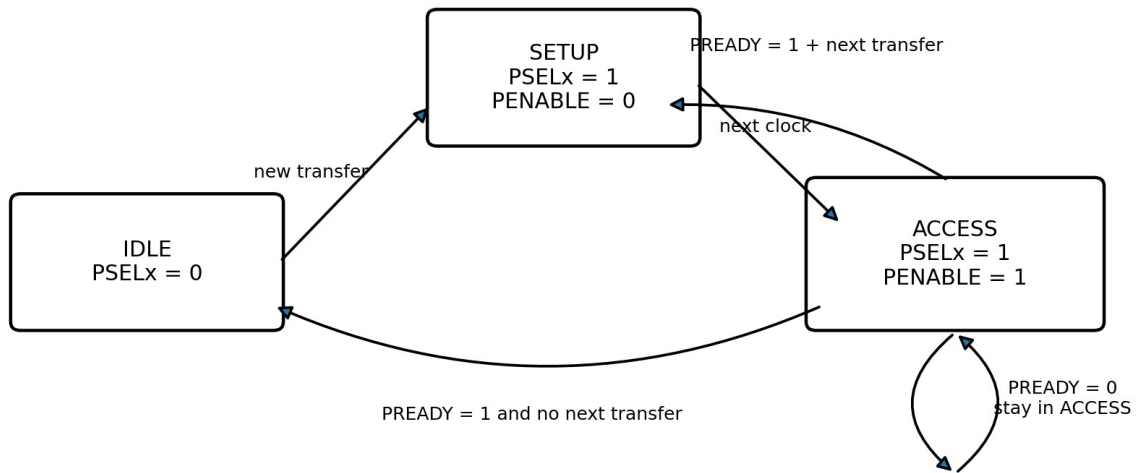
3. Three APB FSM States

APB is synchronous and non-pipelined. A normal transfer needs at least two clock cycles: one SETUP cycle and at least one ACCESS cycle. If the slave is not ready, the ACCESS phase becomes longer.

State	Main signal values	What happens
IDLE	PSELx = 0	No transfer is active. This is the default state.
SETUP	PSELx = 1, PENABLE = 0	Requester puts address/control/data on the bus. SETUP lasts exactly one cycle.
ACCESS	PSELx = 1, PENABLE = 1	The transfer is active. The FSM waits for PREADY from the Completer.

The picture below is my simplified view of the state machine:

APB State Machine - Simple Student View



4. APB State Transition Table

Current state	PSELx	PENABLE	PREADY	Condition	Next state
IDLE	0	0	X	No new transfer	IDLE
IDLE	-> 1	0	X	New transfer starts	SETUP
SETUP	1	0	X	After exactly one clock cycle	ACCESS
ACCESS	1	1	0	Completer is not ready	ACCESS
ACCESS	1	1	1	Transfer finishes, no next transfer	IDLE
ACCESS	1	1	1	Transfer finishes, another transfer follows	SETUP

Here X means “do not care for this transition.” The important point is that PREADY only controls leaving ACCESS. PREADY = 0 means keep waiting in ACCESS. PREADY = 1 means the current transfer completes.

5. Important RTL / Design Verification Point

This is one of the most important rules for me when writing RTL or a testbench: the transaction information must stay stable from SETUP to ACCESS, and it must also stay stable during every wait-state cycle when PREADY = 0.

- PADDR
- PPROT (if present)
- PWRITE
- PWDATA for a write
- PSTRB (if present)

Cycle:	T1	T2	T3	T4
State:	SETUP	ACCESS	ACCESS	ACCESS
PSELx	1	1	1	1
PENABLE	0	1	1	1
PREADY	X	0	0	1
PADDR	A	A	A	A
PWRITE	W	W	W	W
PWDATA	D	D	D	D

What I would check in DV: if PREADY stays LOW, the requester must not change the address, write/read direction, write data, strobe, or protection information. If any of these change before the transfer completes, I would consider it a protocol violation.

6. Quick Interview Recap

The shortest sentence I use to remember APB is: **IDLE: PSEL=0 -> SETUP: PSEL=1, PENABLE=0 -> ACCESS: PSEL=1, PENABLE=1, and ACCESS only finishes when PREADY=1.**

Another simple idea: PSEL starts the transfer, PENABLE starts the access phase, PREADY finishes the transfer. PRDATA and PSLVERR are only meaningful at the final completed access. This is how I currently understand the core APB4 handshake.

Source used: ARM IHI 0024E, AMBA APB Protocol Specification (especially Chapter 2, Chapter 4, Appendix A and Appendix B).